PCET'S
Pimpri Chinchwad College of Engineering

Introduction to R

Faculty of Mathematics

Department of First Year Engineering

What is R?

- R is an environment for data manipulation, statistical computing, graphics display and data analysis.
- Effective data handling and storage of outputs is possible.
- Simple as well as complicated calculations are possible.
- Graphical display on-screen and hardcopy are possible.
- Programming language is effective which includes all possibilities just like any other good programming language.

Switching to R?

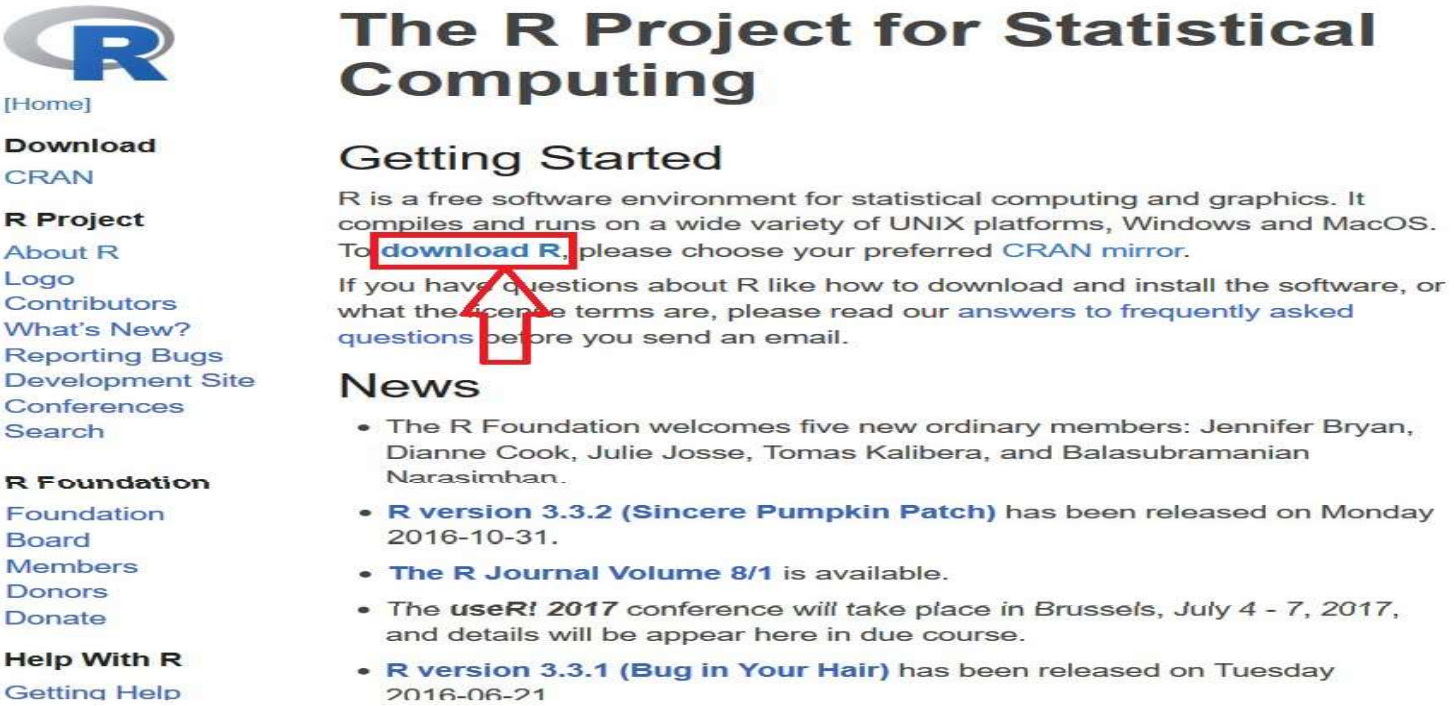
- Many people, research and design offices, analytical firms have started using R.
- R is a free software.
- It has a computer language which is convenient to use for statistical and graphical applications.
- Many statistical packages are freely available. So you may also consider switching to R.
- The commands can be saved, run and stored in script files.

Switching to R?

- R is available for Windows, Unix, Linux and Macintosh platforms.
- Built in and contributed packages are available, and users are provided tools to make packages. It is possible to contribute own packages.
- This language provides the logical control of branching and looping, and modular programming using functions.
- Error messages are provided while executing a programme or a function which are helpful in finding the errors.
- R is an interpreted computer language (Not compiler).

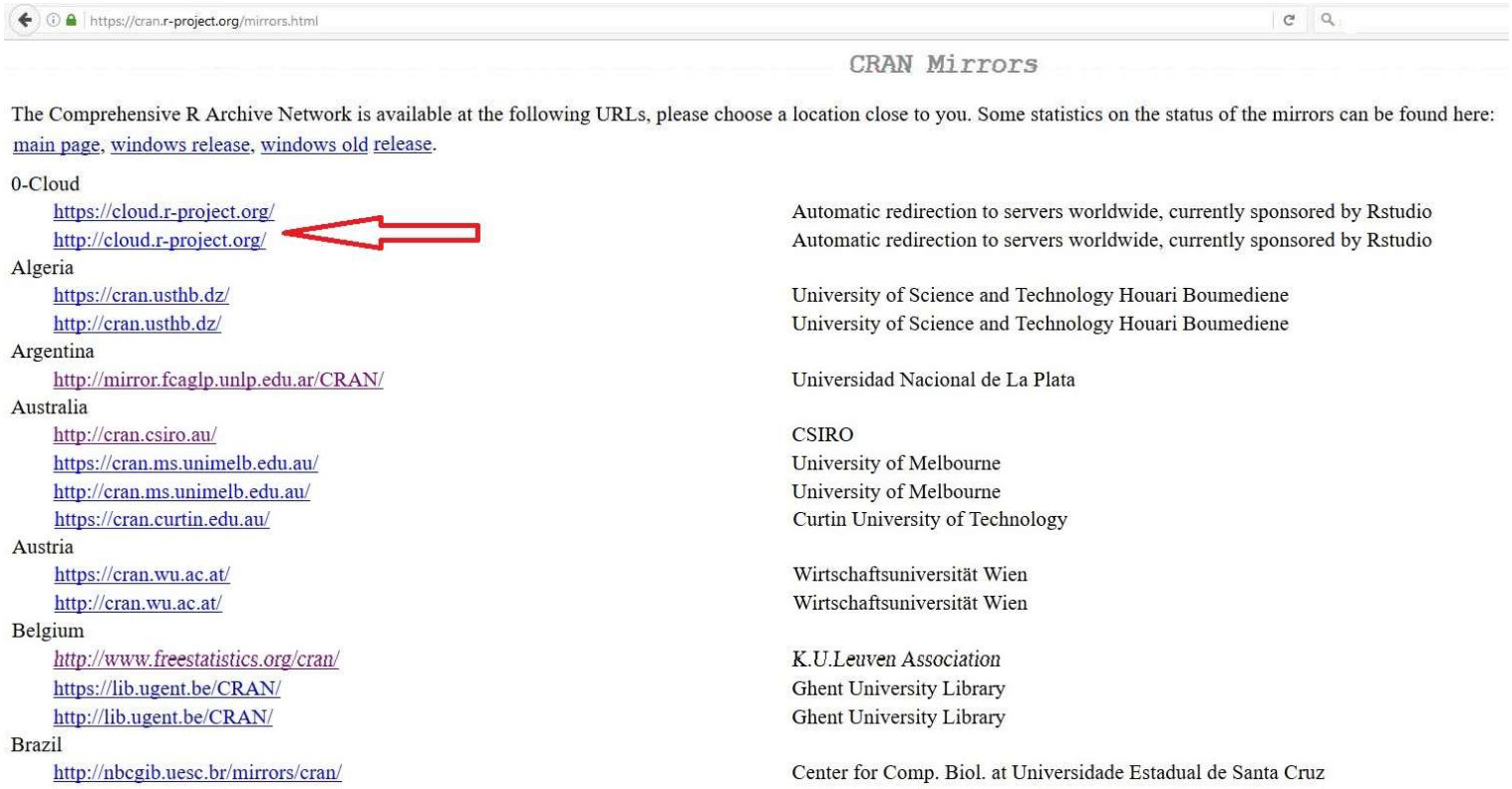
Installing R:

- You may install R in a windows or Apple computer by downloading from **https://www.r-project.org**
- Click on **download R**



Installing R:

- Choose any mirror and click on the link.



https://cran.r-project.org/mirrors.html

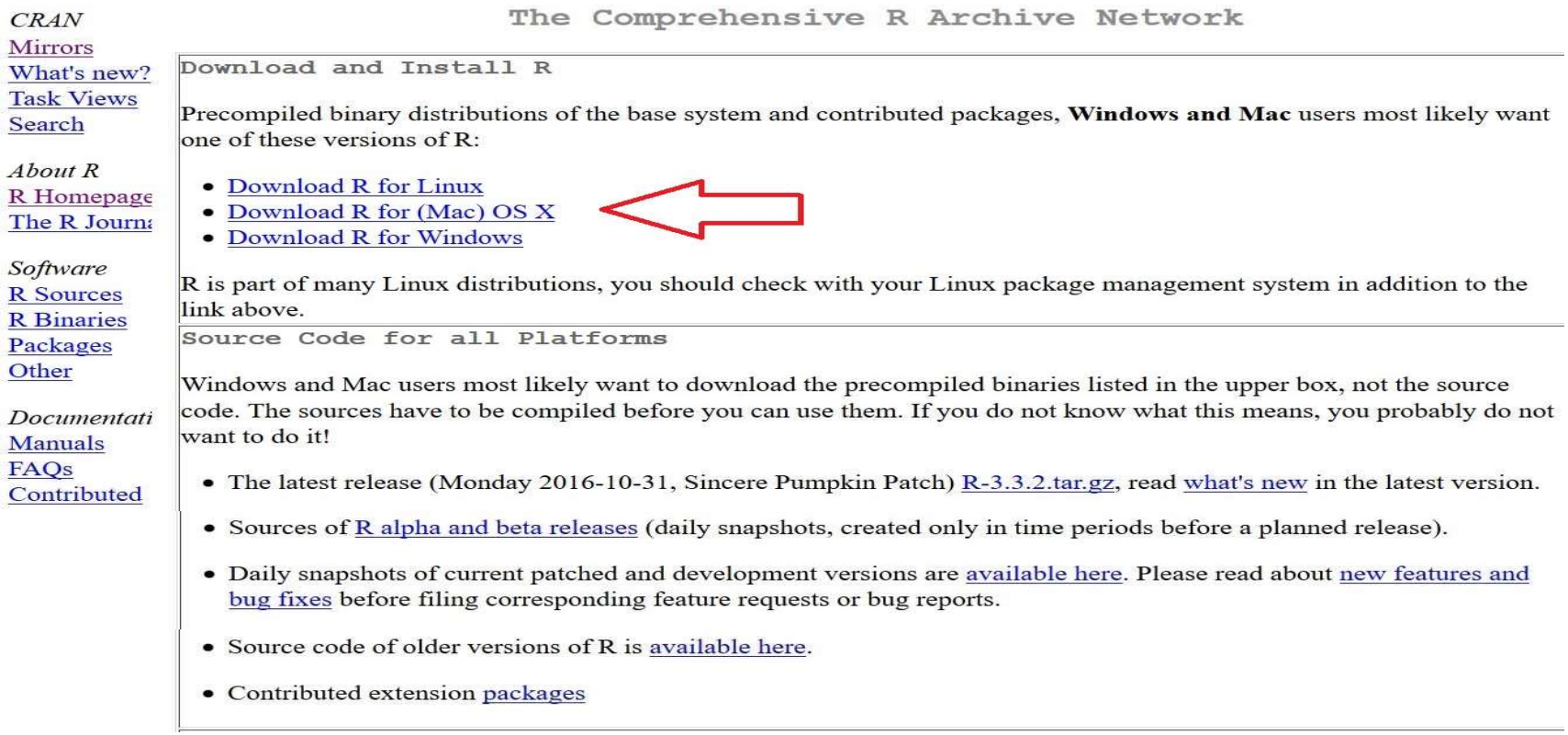
CRAN Mirrors

The Comprehensive R Archive Network is available at the following URLs, please choose a location close to you. Some statistics on the status of the mirrors can be found here: main page, windows release, windows old release.

0-Cloud	
https://cloud.r-project.org/	Automatic redirection to servers worldwide, currently sponsored by Rstudio
http://cloud.r-project.org/	Automatic redirection to servers worldwide, currently sponsored by Rstudio
Algeria	
https://cran.usthb.dz/	University of Science and Technology Houari Boumediene
http://cran.usthb.dz/	University of Science and Technology Houari Boumediene
Argentina	
http://mirror.fcaglp.unlp.edu.ar/CRAN/	Universidad Nacional de La Plata
Australia	
http://cran.csiro.au/	CSIRO
https://cran.ms.unimelb.edu.au/	University of Melbourne
http://cran.ms.unimelb.edu.au/	University of Melbourne
https://cran.curtin.edu.au/	Curtin University of Technology
Austria	
https://cran.wu.ac.at/	Wirtschaftsuniversität Wien
http://cran.wu.ac.at/	Wirtschaftsuniversität Wien
Belgium	
http://www.freestatistics.org/cran/	K.U.Leuven Association
https://lib.ugent.be/CRAN/	Ghent University Library
http://lib.ugent.be/CRAN/	Ghent University Library
Brazil	
http://nbcgib.uesc.br/mirrors/cran/	Center for Comp. Biol. at Universidade Estadual de Santa Cruz

Installing R:

- Else, download it from the Comprehensive R Archive Network (CRAN) website: **http://cran.r-project.org/**

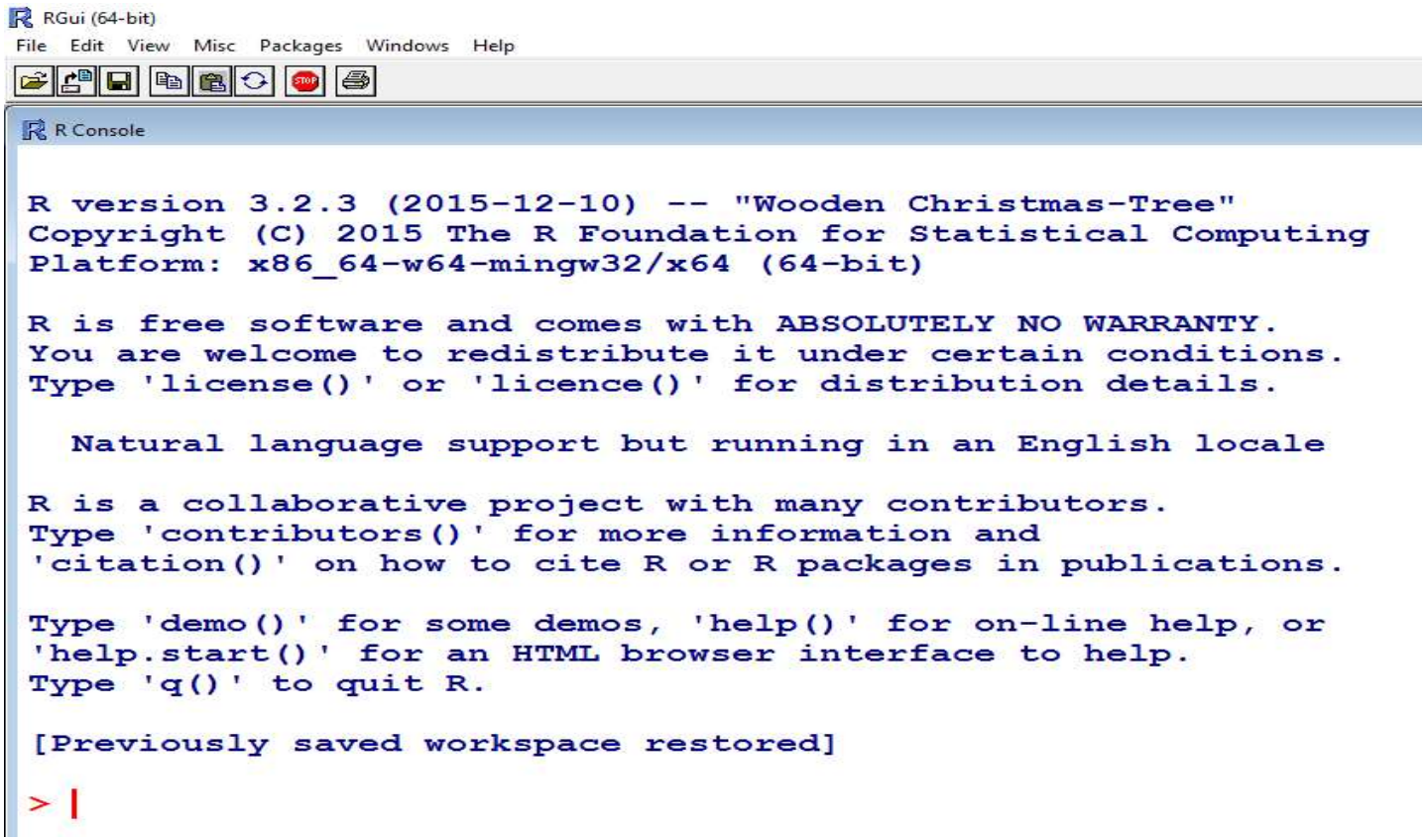


Starting with R:

To start R, double click on the icon 

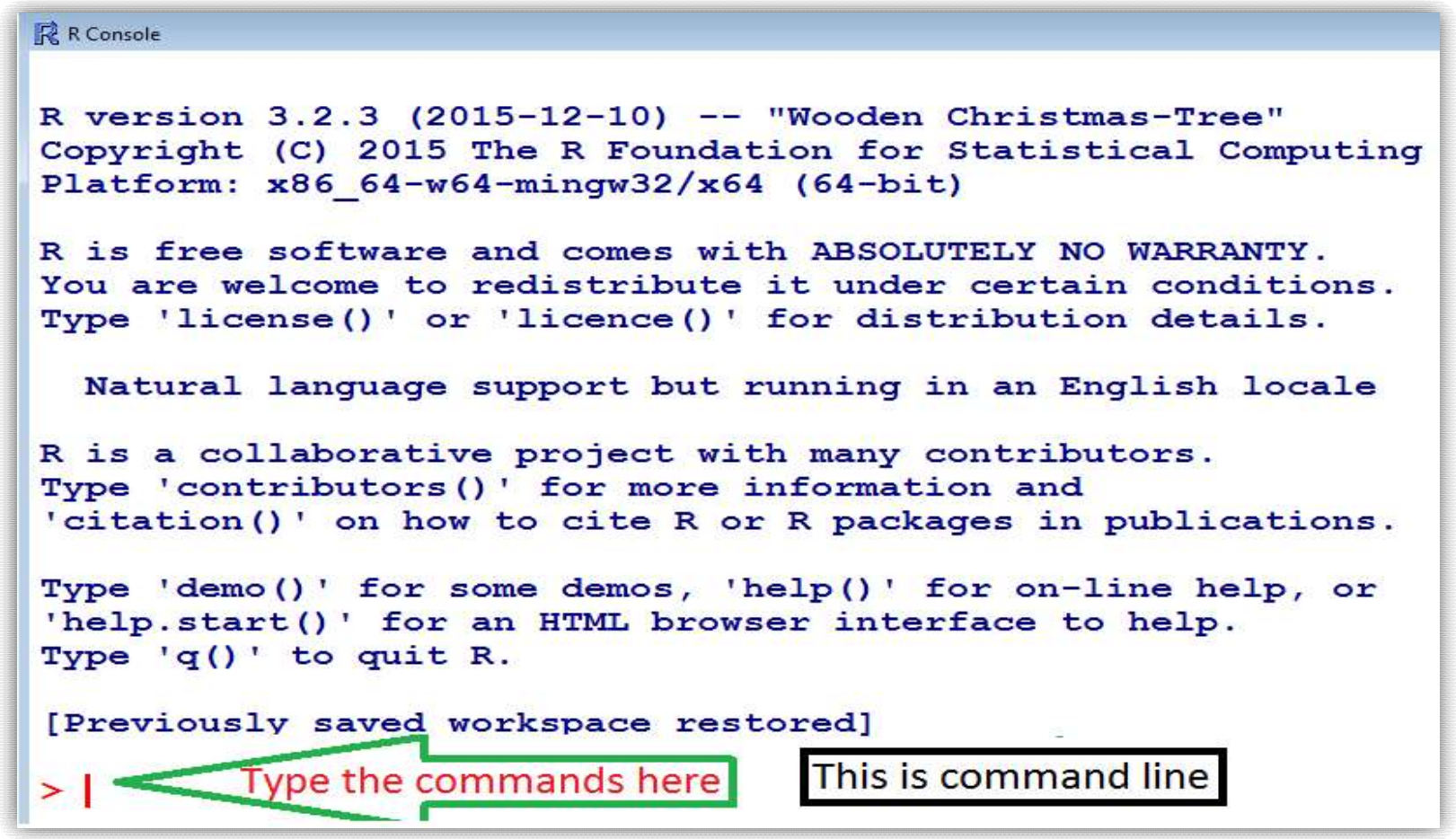
.

Then we get the following Gui (Graphic user interface) window screen



Command Line versus Scripts

What is command line?



Command Line versus Scripts

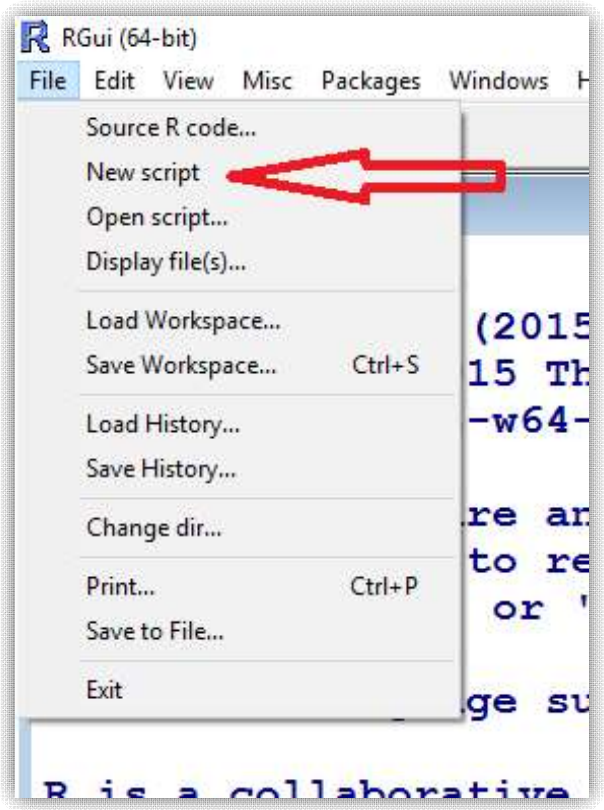
What is command line?

- Execution of commands in R is not menu driven. (Not like Clicking over buttons to get outcome).
- We need to type the commands.
- Single line and multi line commands are possible to write.
- When writing multi-line programs, it is useful to use a text editor rather than execute everything directly at the command line.

Command Line versus Scripts

Option 1:

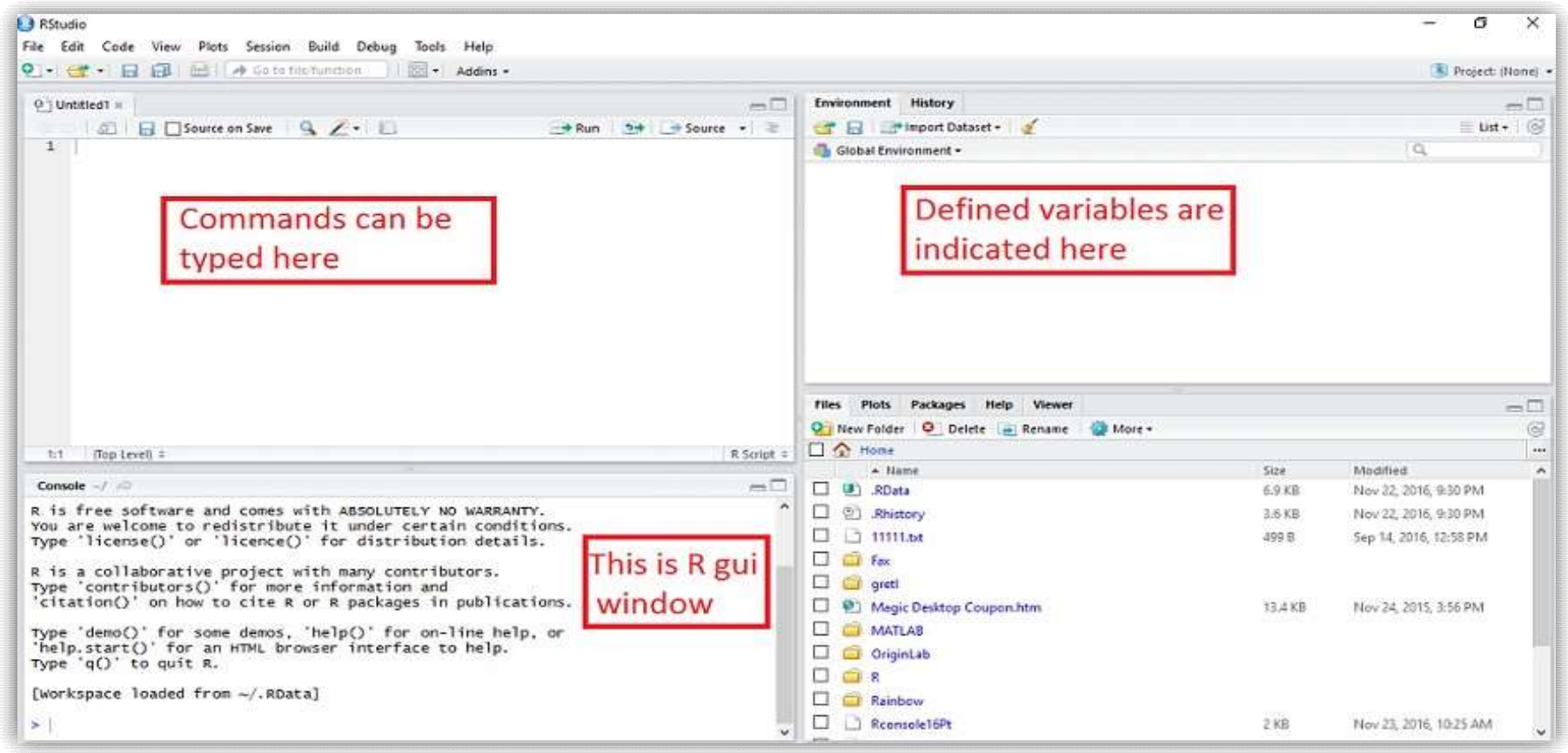
- One may use R's own built-in editor.
- It is accessible from the **RGui** menu bar. Click File and then click on New script.
- At this point R will open a window entitled Untitled-R Editor.
- We may type and edit in this.
- If we want to execute a line or a group of lines, just highlight them and press **Ctrl+R**.



Command Line versus Scripts

Option 2:

Use R studio software



Installing R Studio:

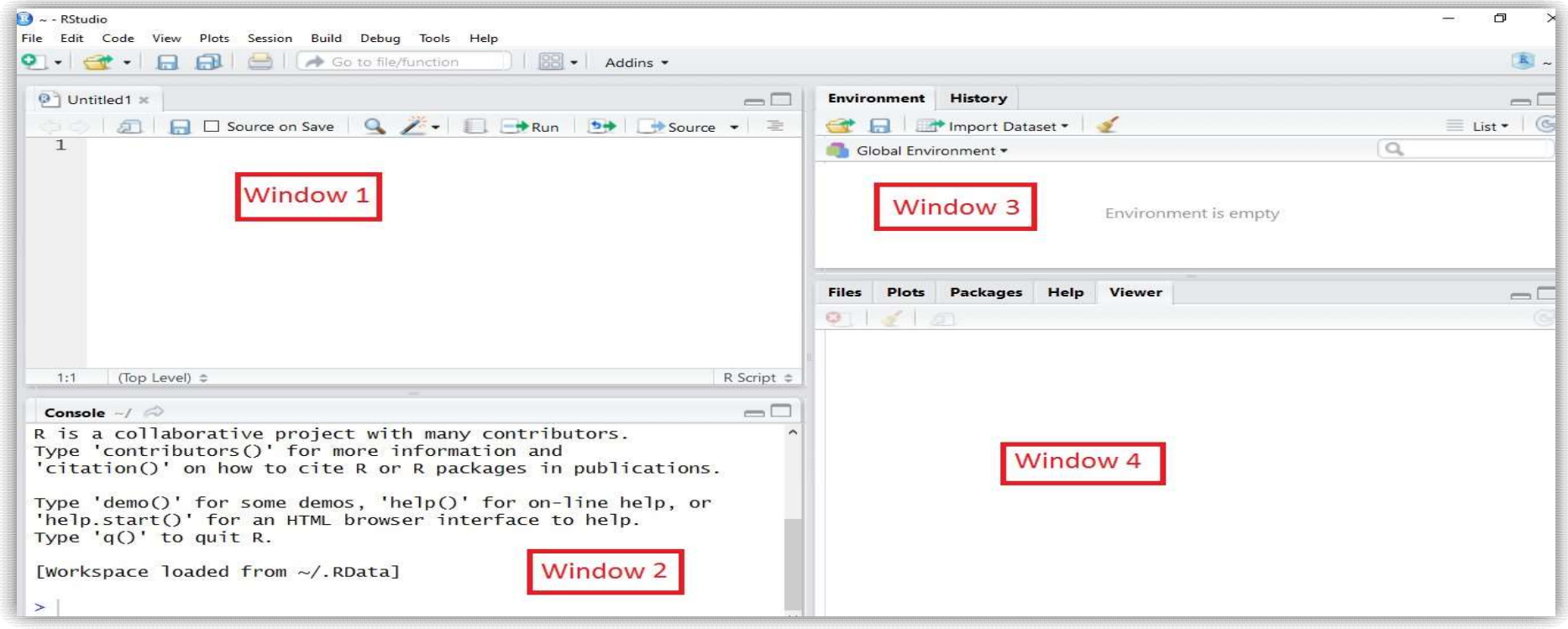
- Rstudio is an editing software which helps in running the R software.
- Several such editors are available, e.g. Tinn R (https://sourceforge.net/projects/tinn-r)
- Rstudio is written in C++ programming language.
- Like R, Rstudio is also free and open-source integrated development environment (IDE) for R.
- Download R-Studio software from website https://www.rstudio.com/

Installing R Studio:



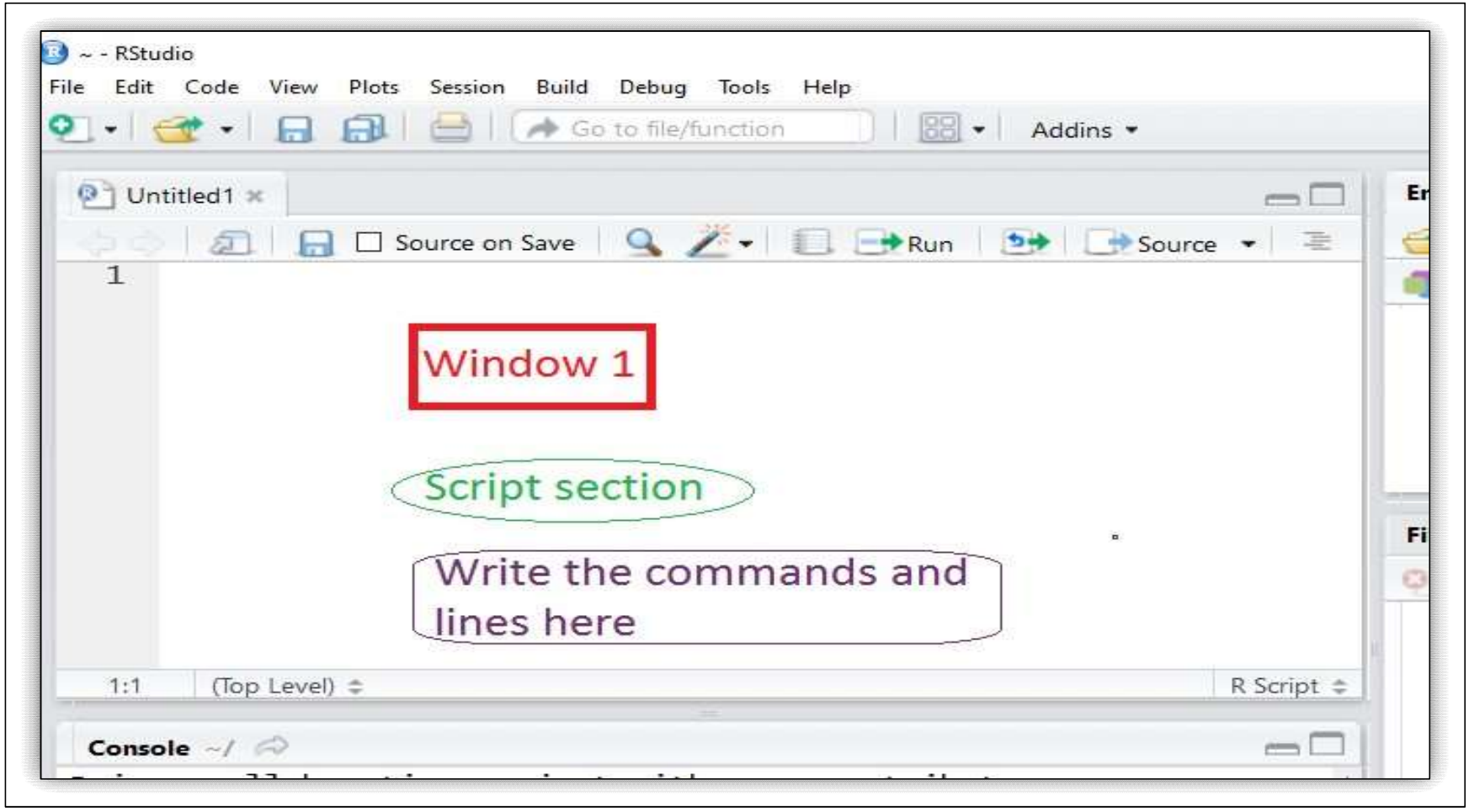
Introduction to R Studio

- It is an interface between R and us.
- More useful for beginners. It makes coding easier.
- When we start R studio, we see 4 windows.



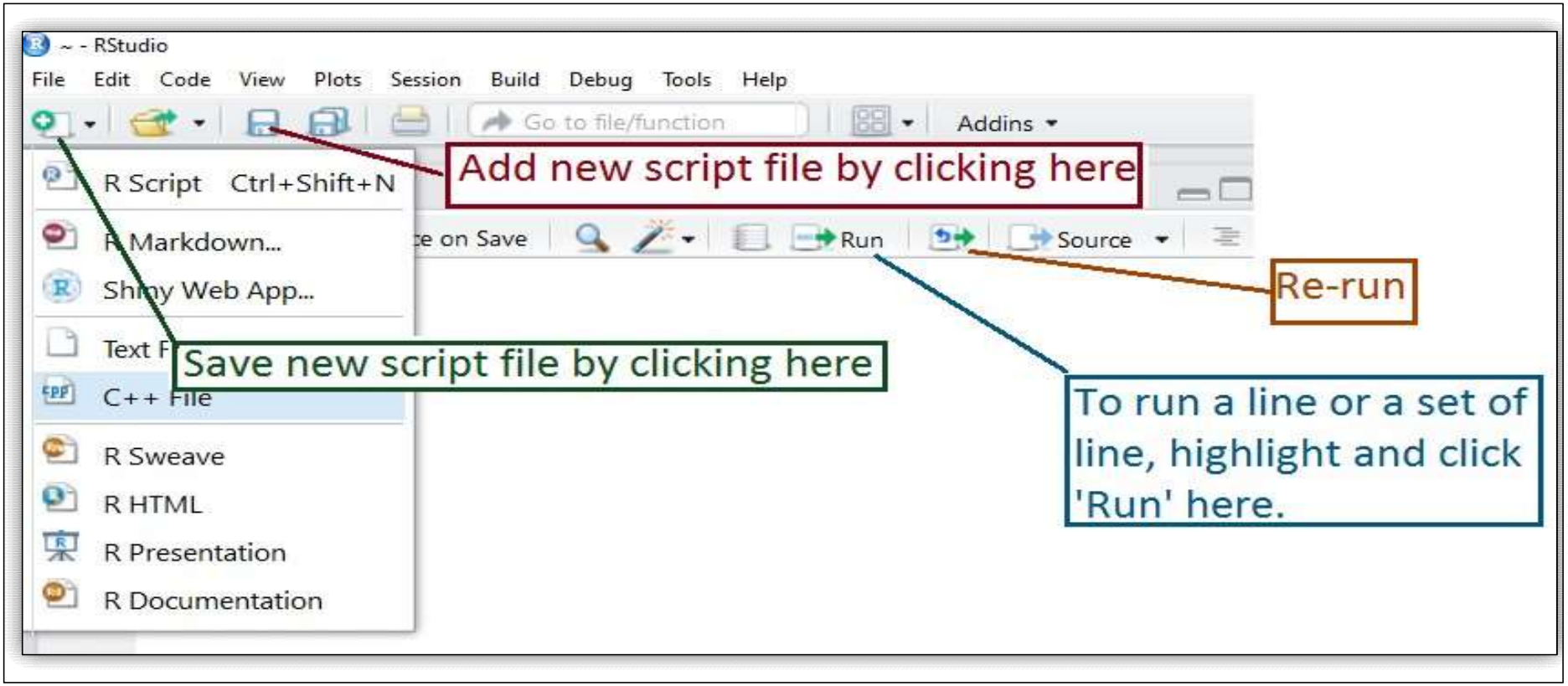
Introduction to R Studio

Description of Window 1



Introduction to R Studio

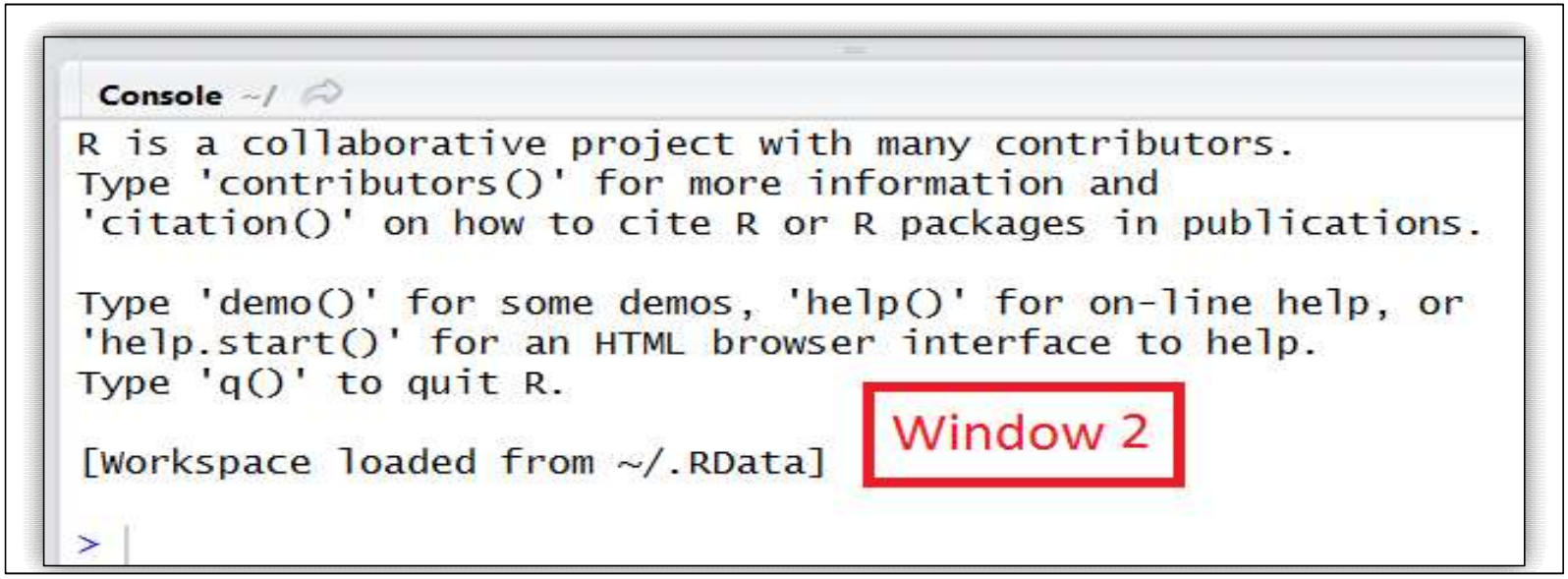
Description of Window 1



Introduction to R Studio

Description of Window 2: Console

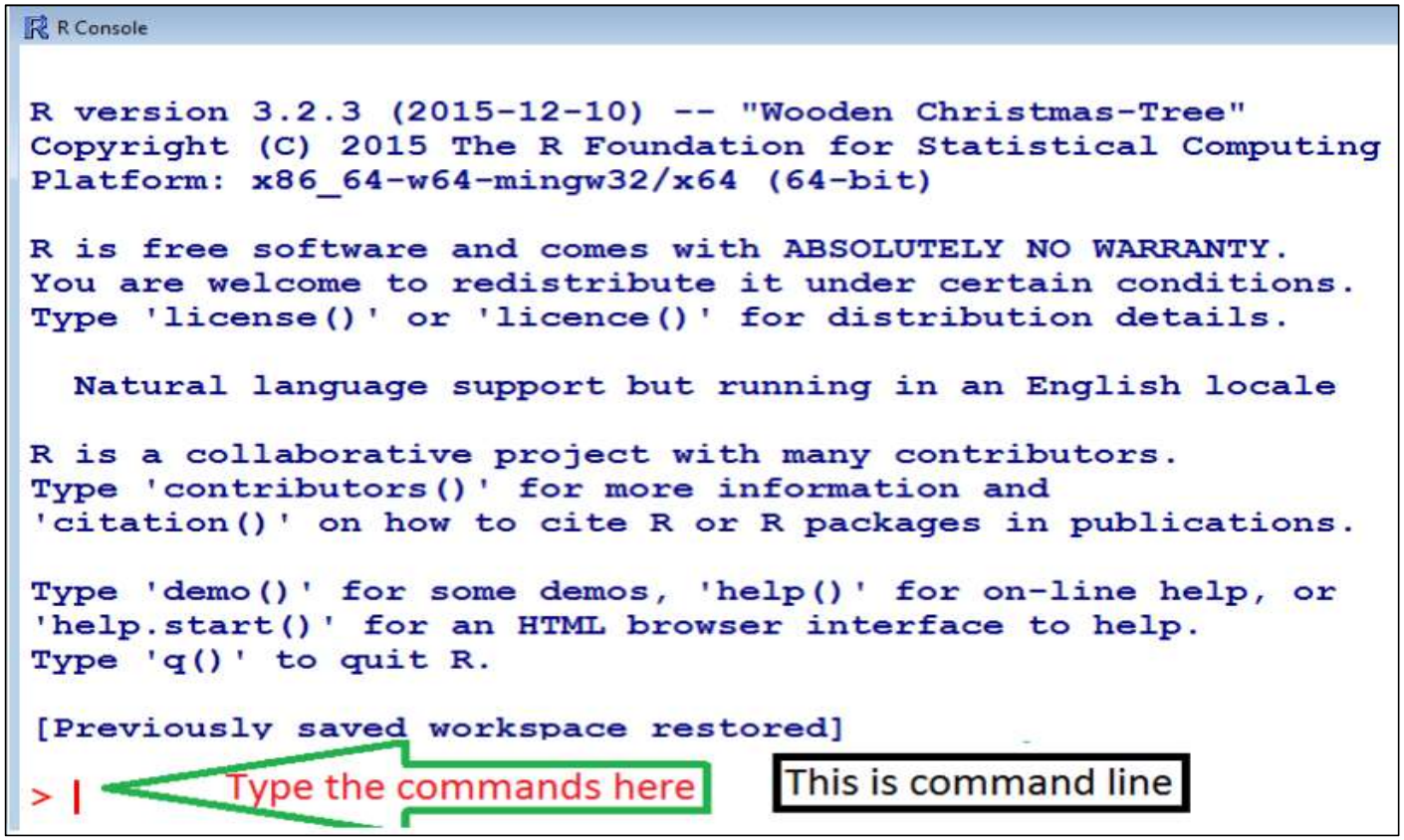
- R program window appears here.
- Calculations take place in console window.
- One can write program in console also but it is hard to make corrections and experiments with the coding



Introduction to R Studio

Description of Window 2: Console

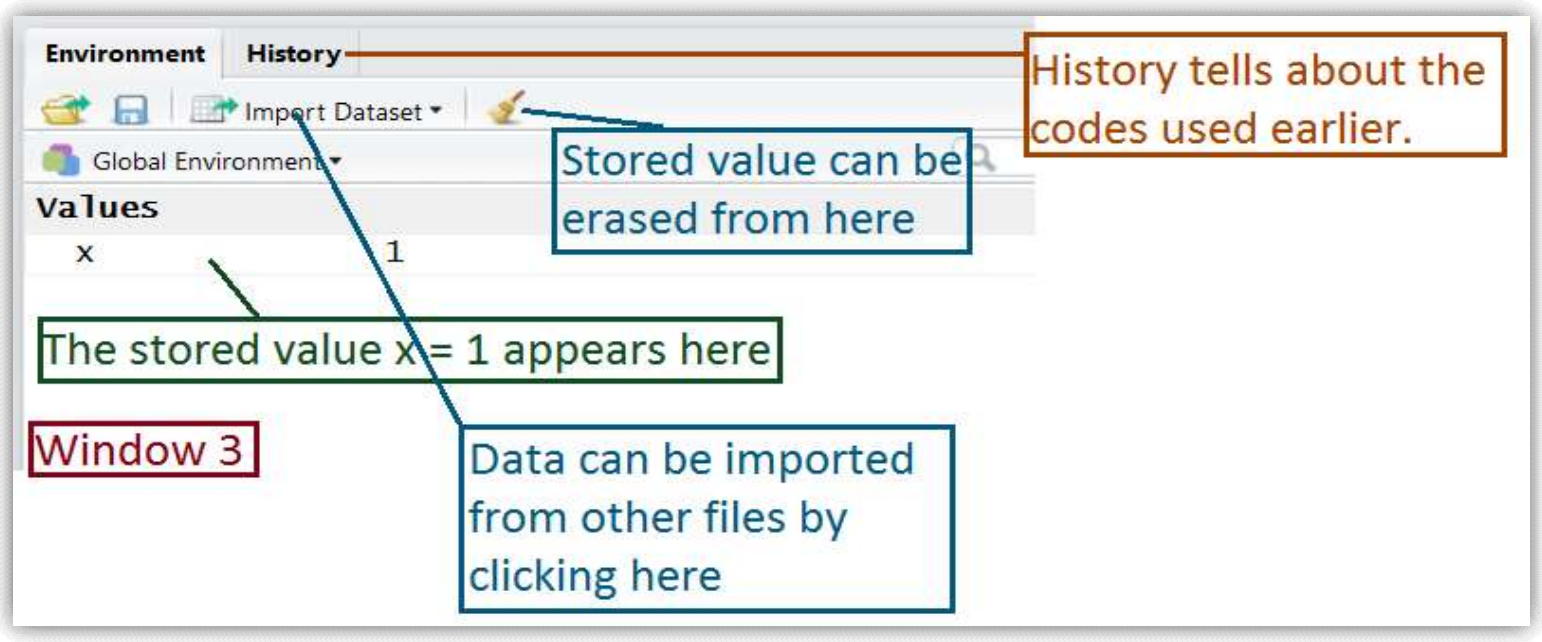
- Press the up-arrow key to recall commands and edit them.
- Use the Esc (Escape) key to cancel a command.



Introduction to R Studio

Description of Window 3: Environment window

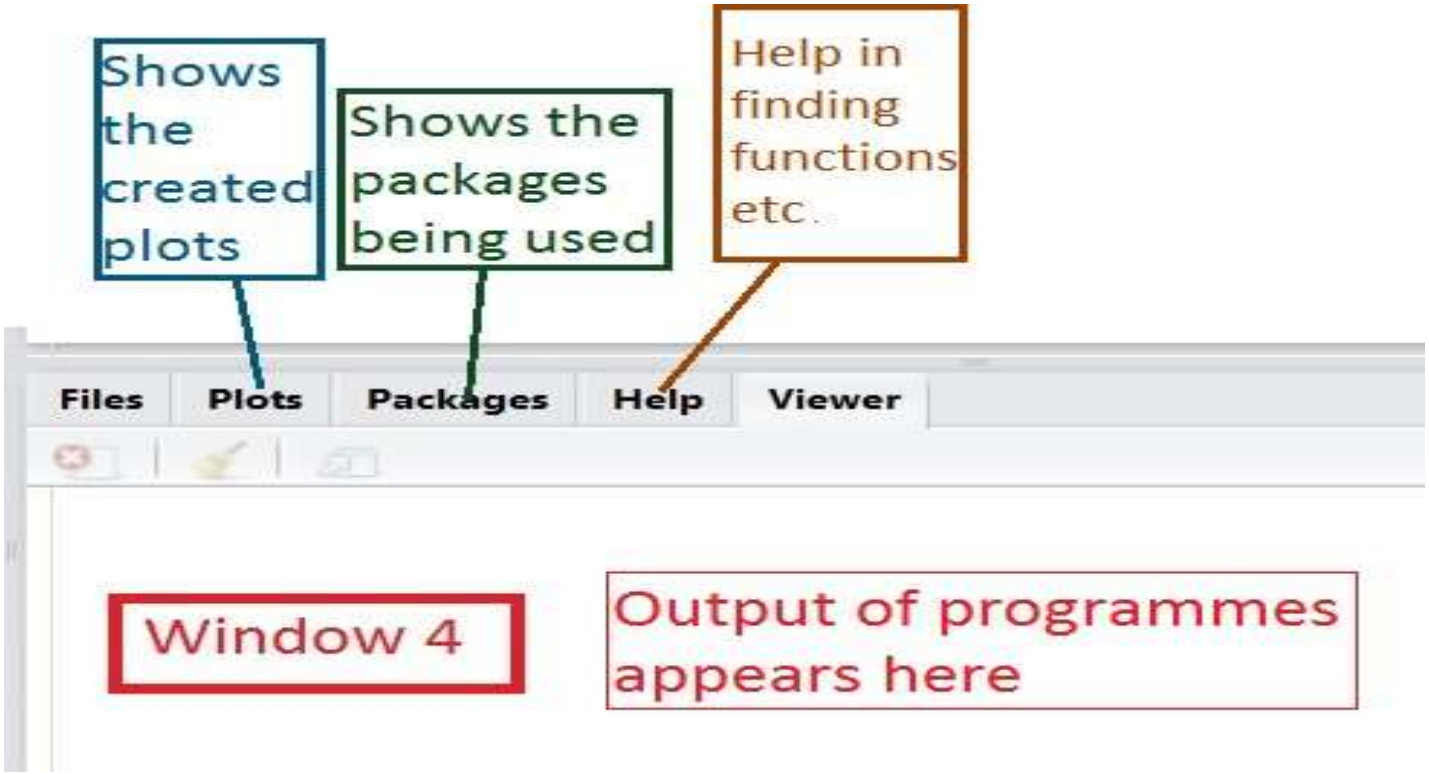
All the variables and objects used in the program appear here. The nature and values of variables and objects also appear here.



Introduction to R Studio

Description of Window 4: Output window

The output of programs appears in this window.



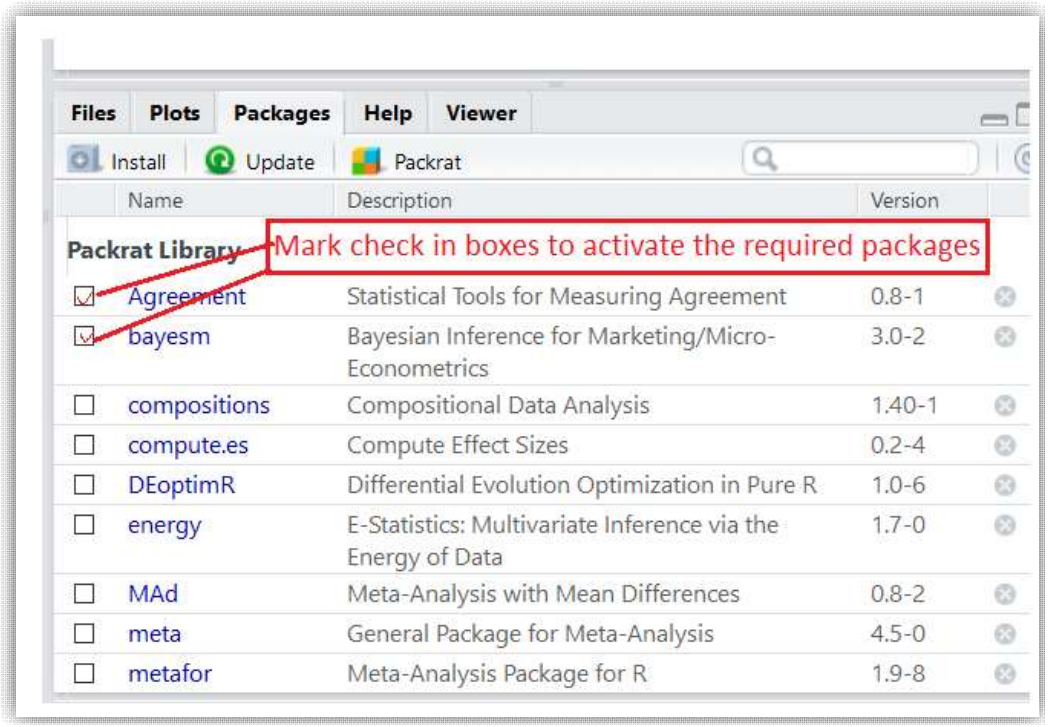
Introduction to R Studio

Description of Window 4: Output window

Packages:

All the packages being installed appear here.

Those packages which are not active, check mark in the boxes to activate them.



Introduction to R Studio

Description of Window 4: Output window

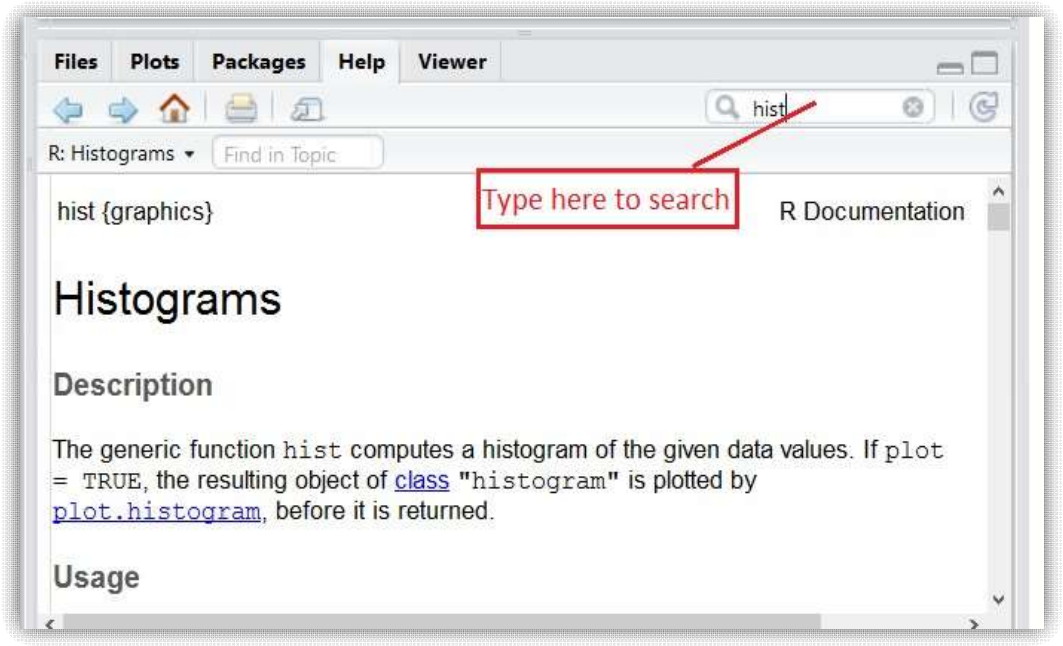
Help:

Various types of help can be asked.

E.g., to know about histogram,

type hist and all the Information

About histogram appears.



Introduction to R Studio

Description of Window 4: Output window

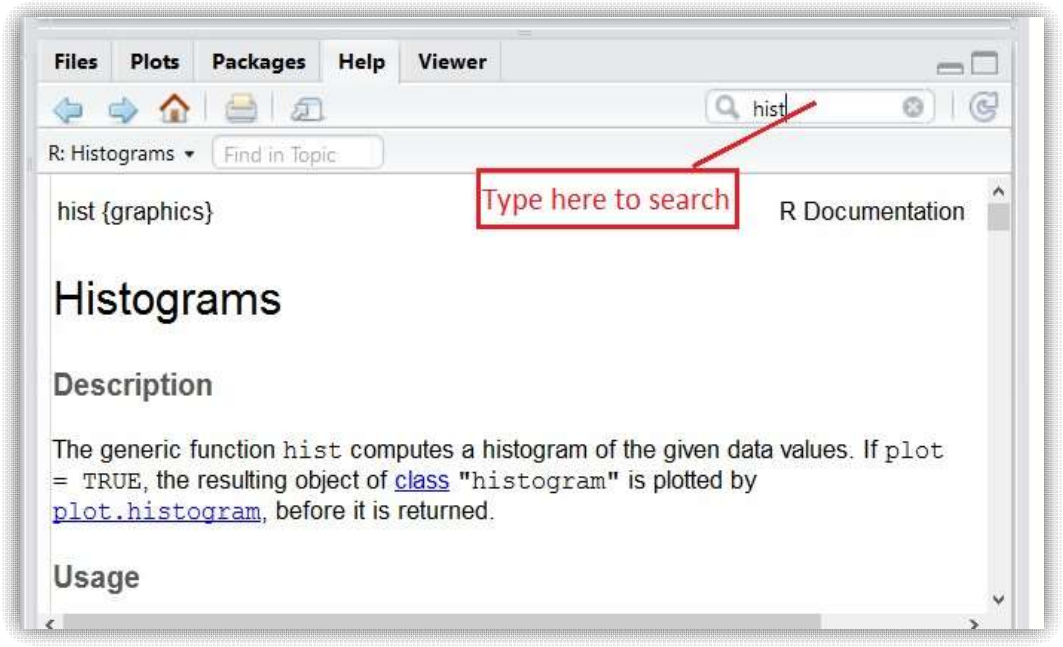
Help:

Various types of help can be asked.

E.g., to know about histogram,

type hist and all the Information

About histogram appears.



Libraries in R

- R provides many functions and one can also write own. Functions and datasets are organised into libraries
- To use a library, simply type the `library` function with the name of the library in brackets.

```
library(.)
```

- For example, to load the `spatial` library type:

```
library(spatial)
```

Libraries in R

Examples of libraries that come as a part of base package in R.

`MASS :` package associated with Venables and Ripley's book entitled Modern Applied Statistics using S-Plus.

`mgcv :` generalized additive models.

Contents of Libraries

It is easy to use the `help` function to discover the contents of library packages.

Here is how we find out about the contents of the `spatial` library:

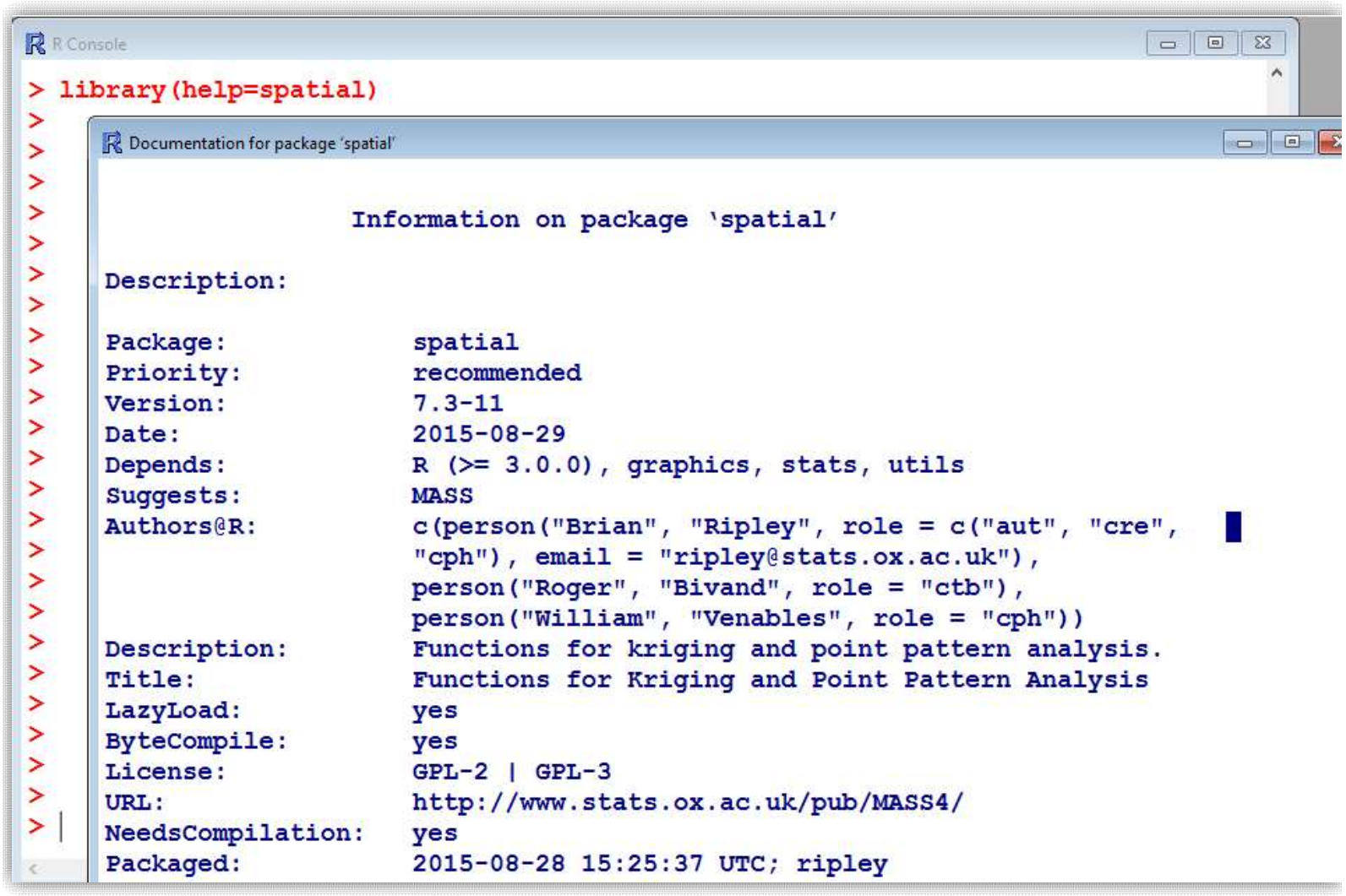
`library(help=spatial)` returns

Output:

```
      Information on package 'spatial' Description:
Package:  spatial
Priority: recommended Version:7.3-8
```

followed by a list of all the functions and data sets.

Then we get....


R Console
> library(help=spatial)
Documentation for package 'spatial'
Information on package 'spatial'
Description:
Package: spatial
Priority: recommended
Version: 7.3-11
Date: 2015-08-29
Depends: R (>= 3.0.0), graphics, stats, utils
Suggests: MASS
Authors@R: c(person("Brian", "Ripley", role = c("aut", "cre",
"cph"), email = "ripley@stats.ox.ac.uk"),
person("Roger", "Bivand", role = "ctb"),
person("William", "Venables", role = "cph"))
Description: Functions for kriging and point pattern analysis.
Title: Functions for Kriging and Point Pattern Analysis
LazyLoad: yes
ByteCompile: yes
License: GPL-2 | GPL-3
URL: http://www.stats.ox.ac.uk/pub/MASS4/
NeedsCompilation: yes
Packaged: 2015-08-28 15:25:37 UTC; ripley

Installing Packages and Libraries

The base R package contains programs for basic operations.

It does not contain some of the libraries necessary for advanced statistical work.

Specific requirements are met by special packages.

They are downloaded and their downloading is very simple.

Installing Packages and Libraries

To install any package,

- run the R program,
- Then on the command line, use the `install.packages` function to download the libraries we want.

Examples:

- The package `rmeta` contains the statistical tools for meta analysis.
- The package `Agreement` contains statistical tools for measuring agreement.

The packages `rmeta` or `Agreement` can be installed by

```
install.packages("rmeta")
install.packages("Agreement")
```

Then we get…

Installing Package “rmeta”



11 Most Important Packages in R for Data Science

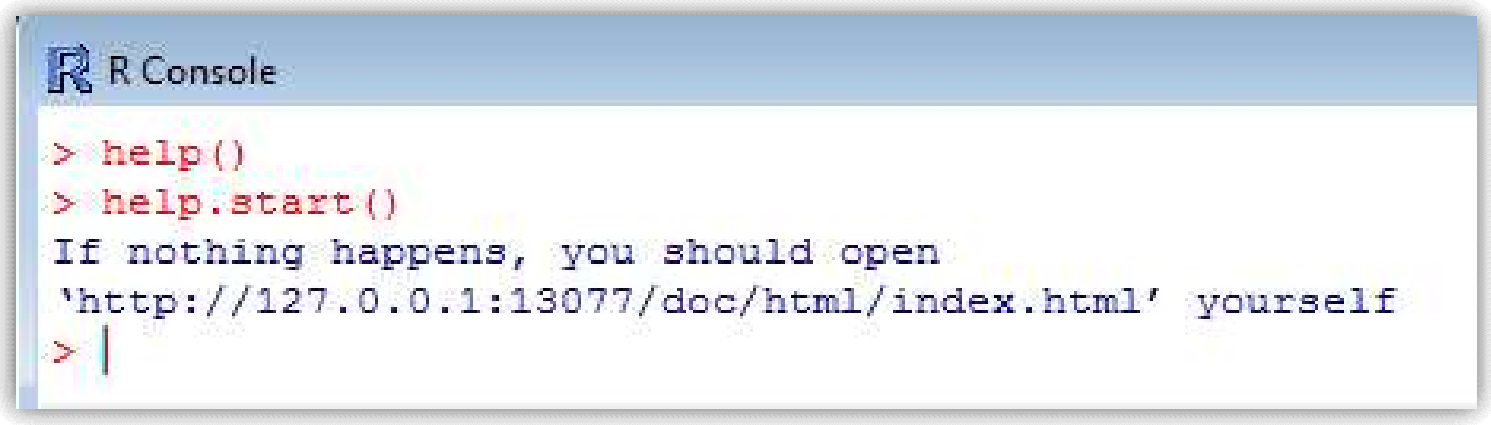
1. ggplot2
2. dplyr
3. tidyr
4. Caret [Just introduction]
5. Tidyverse
6. Tigerstat
7. Descstat
8. Grid

R is beyond these packages. You can deep dive to explore vast packages of the R.

Getting Help in R

`'help()'` for on-line help,

or `'help.start()'` for an HTML browser interface to help.



then we get....

127.0.0.1:13077/doc/html/index.html

Statistical Data Analysis



Manuals

An Introduction to R
Writing R Extensions
R Data Import/Export
The R Language Definition
R Installation and Administration
R Internals

Reference

Packages
Search Engine & Keywords

Miscellaneous Material

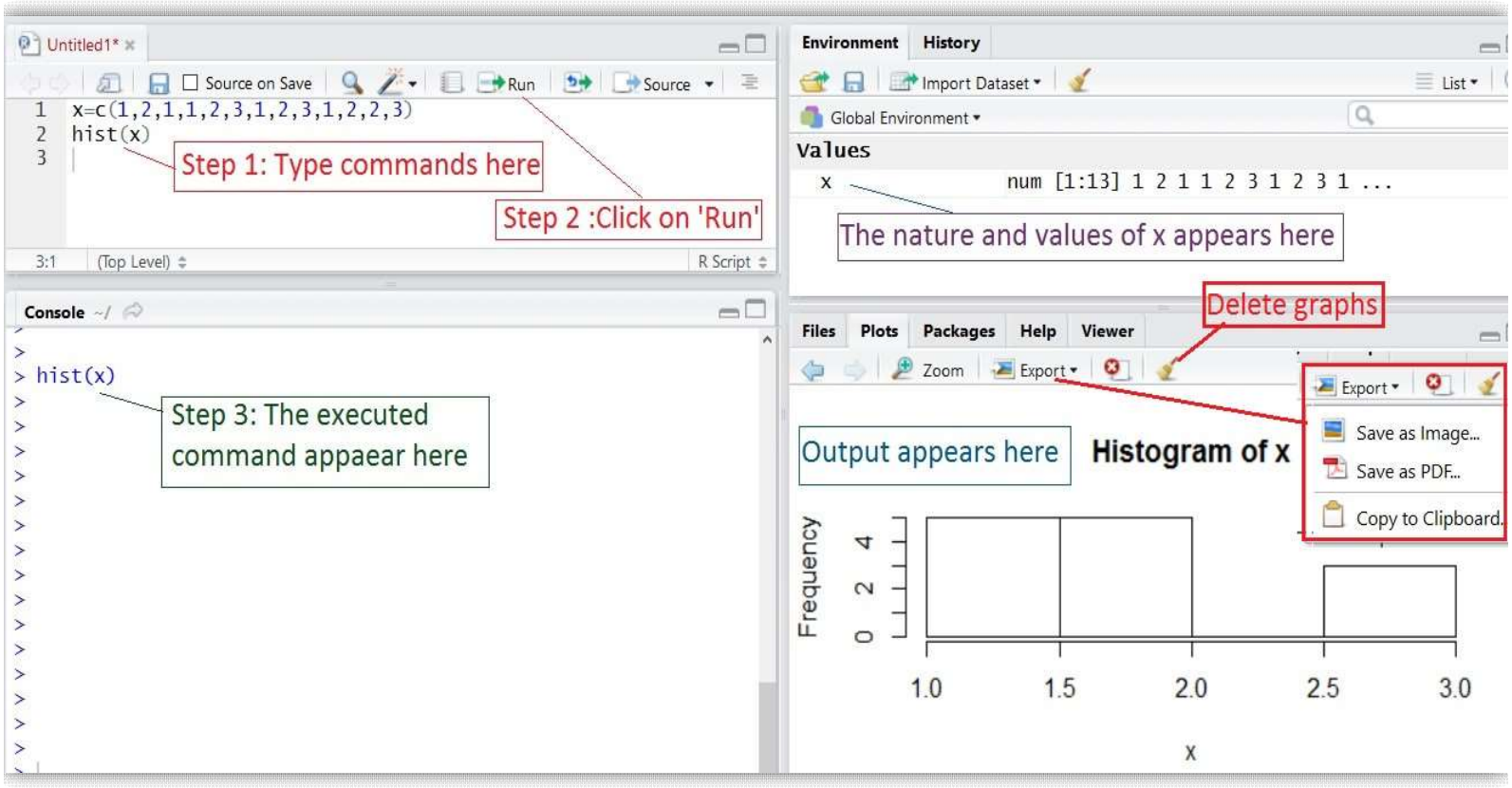
About R
License
NEWS
Authors
Frequently Asked Questions
User Manuals
Resources
Thanks
Technical papers

Material specific to the Windows port

CHANGES up to R 2.15.0
Windows FAQ

Example:

Histogram of values 1,2,1,1,2,3,1,2,3,1,2,2,3 R studio has following operation and output:



Cleaning up the Windows

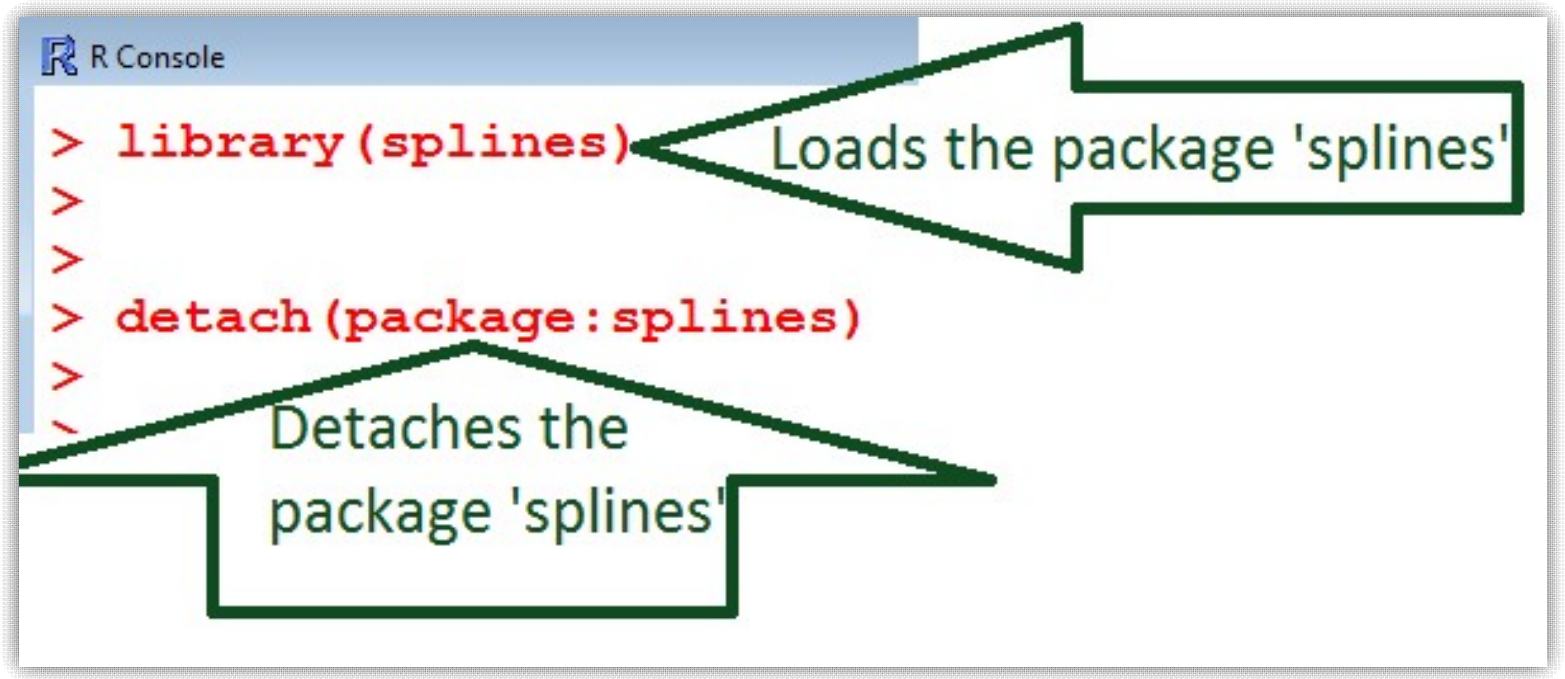
- We assign names to variables when analyzing any data.
- It is good practice to remove the variable names given to any data frame at the end each session in R.
- This way, variables with same names but different properties will not get in each others way in subsequent work.
- `rm()` command removes variable names.
- For example, `rm(x,y,z)` removes the variables x, y and z.

Cleaning up the Windows

- `detach()` command detaches objects from the Search Path.
- It removes it from the `search()` path of available R objects.
- Usually this is either a `data.frame` which has been attached or a package which was attached by `library`.
- To get rid of everything, including data frames, type
- `rm(list=ls())`

Then we get….

Cleaning up the Windows



Conclusion

RStudio is a must-know tool for everyone who works with the R programming language. It's used in data analysis to import, access, transform, explore, plot, and model data, and for machine learning to make predictions on data.

Explore more about R and R-studio……

Everything You Need to Know: R and/or R Studio



Questions

1. What is R, and what are its main characteristics?
2. What id difference between R and R-studio
3. What are advantages of R?
4. How to import data in R?
5. What is a package in R, and how do you install and load packages?
6. How to create a data frame in R?
7. Explain in detail GUI of R-studio